

Standard Form & Indices

Test 10 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 2^{15} (1)

A2. 3^9 (1)

A3. 10^{12} (1)

A4. $6^1 \times 6^2 = 6^3$ (1) = **216** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 11^2 = 1 \div 121$ (1) = **1/121** (1)

B3. $1 \div 2^3 = 1 \div 8$ (1) = **0.125** (1)

Section C — Writing standard form

C1. 5.7 (1) = **5.7×10^4** (1)

C2. 6.8 (1) = **6.8×10^6** (1)

C3. 9.3 (1) = **9.3×10^{-3}** (1)

C4. 5 (1) = **5×10^{-7}** (1)

Section D — Converting back, and calculating

D1. **4,700,000** (1)

D2. **0.00006** (1)

D3. $3 \times 4 = 12$, $10^6 \times 10^3 = 10^9$ (1) = 12×10^9 (1) adjust to standard form: **1.2×10^{10}** (1)

D4. $3 \div 6 = 0.5$, $10^5 \div 10^2 = 10^3$ (1) = 0.5×10^3 (1) adjust to standard form: **5×10^2** (1)

Section E — Powers to know, and estimating roots

E1. **256** (1)

E2. $9^2=81$, $10^2=100$ = between **9 and 10** (1)

Section F — Laws of indices with algebra

F1. n^{16} (1)

F2. r^{12} (1)

F3. s^{15} (1)

F4. $8 \times 4 = 32$ (1) $a^{2+7} = a^9$ (1) = **$32a^9$**

F5. $5^3 = 125$ (1) $b^{2 \times 3} = b^6$ (1) = **$125b^6$**

Section G — Fractional indices (Higher)

G1. $\sqrt{225} = \mathbf{15}$ (1)

G2. cube root of 8000 = **20** (1)

G3. $\sqrt{16} = 4$ (1) $4^3 = \mathbf{64}$ (1)

G4. cube root of 64 = 4, $4^2 = 16$ (1) $1 \div 16 = \mathbf{1/16}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.